Figure 1 - ZIP Header Mismatch

fig1_zip_header_mismatch.pdf

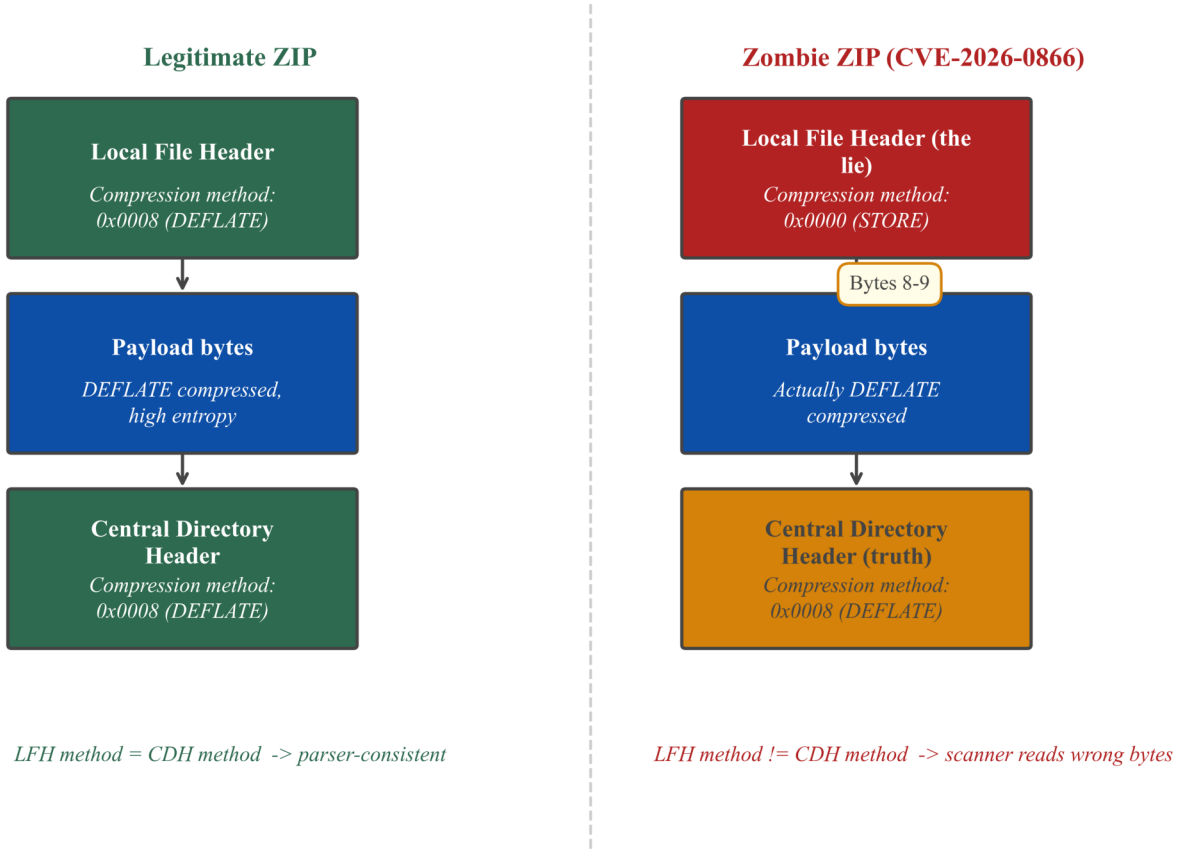

Figure 2 - Attack Taxonomy

fig2_attack_taxonomy.pdf

ID	Variant Name	LFH Method	CDH Method	Payload	Primary Signal
A	Classic Zombie ZIP	STORE (0)	DEFLATE (8)	Compressed	`method_mismatch`
B	Method-only mismatch	DEFLATE (8)	STORE (0)	Stored	`method_mismatch`
C	Gootloader concatenation (real-world dominan	DEFLATE (8)	DEFLATE (8)	Compressed	`eocd_count > 1`
D	Multi-file decoy	STORE (0)	DEFLATE (8)	Compressed	`suspicious_entry_ratio`
E	CRC32 mismatch	DEFLATE (8)	DEFLATE (8)	Compressed	`any_crc_mismatch`
F	Extra field noise	STORE (0)	DEFLATE (8)	Compressed	`structural_combo`
G	High compression mismatch	STORE (0)	DEFLATE (8)	Compressed	`entropy_gap`
H	Size field mismatch	STORE (0)	DEFLATE (8)	Compressed	`size_inconsistency`

Figure 3 - SHAP Importance

fig3_shap_importance.pdf

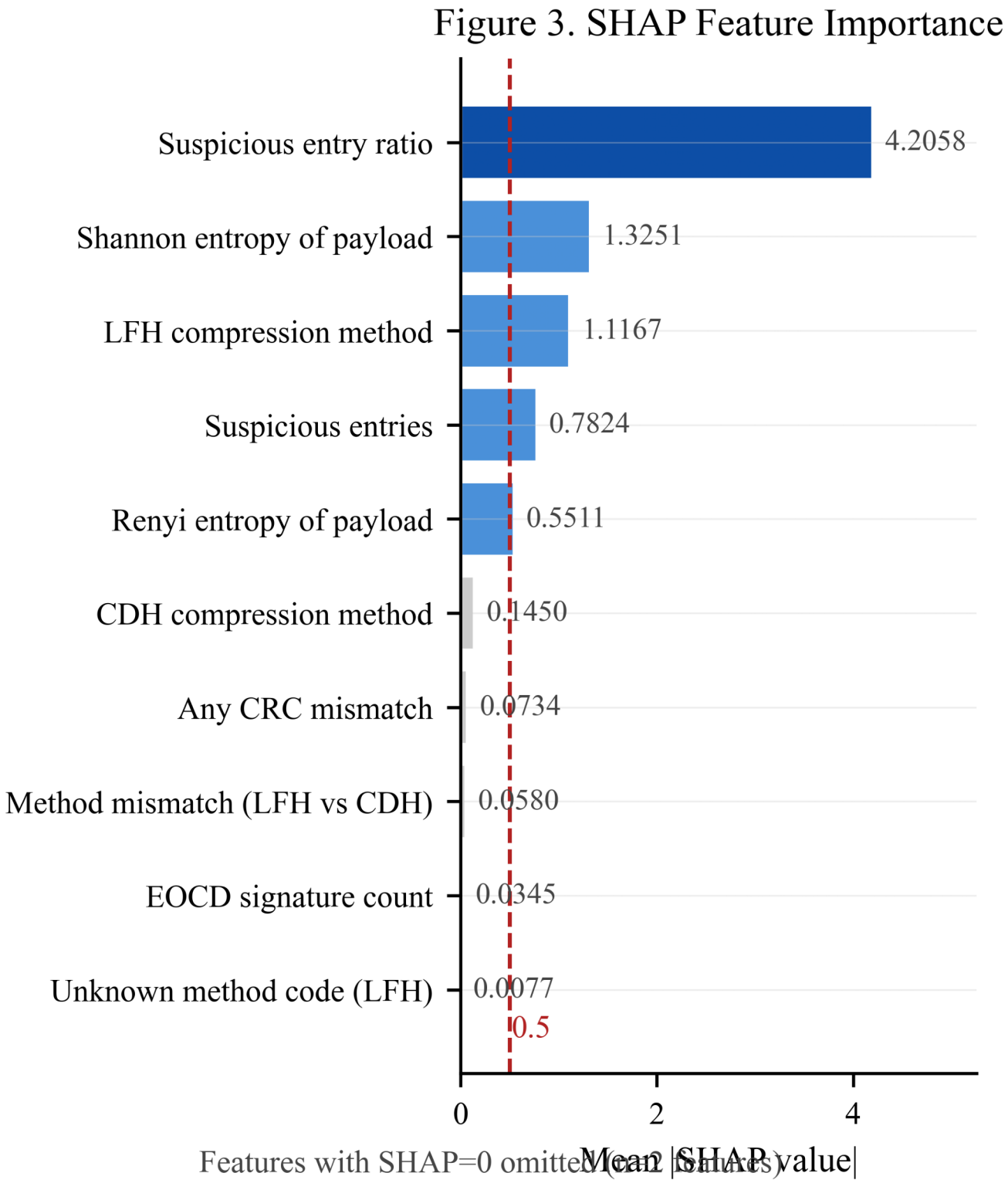

Figure 4 - Generalisation Chart

fig4_generalisation_chart.pdf

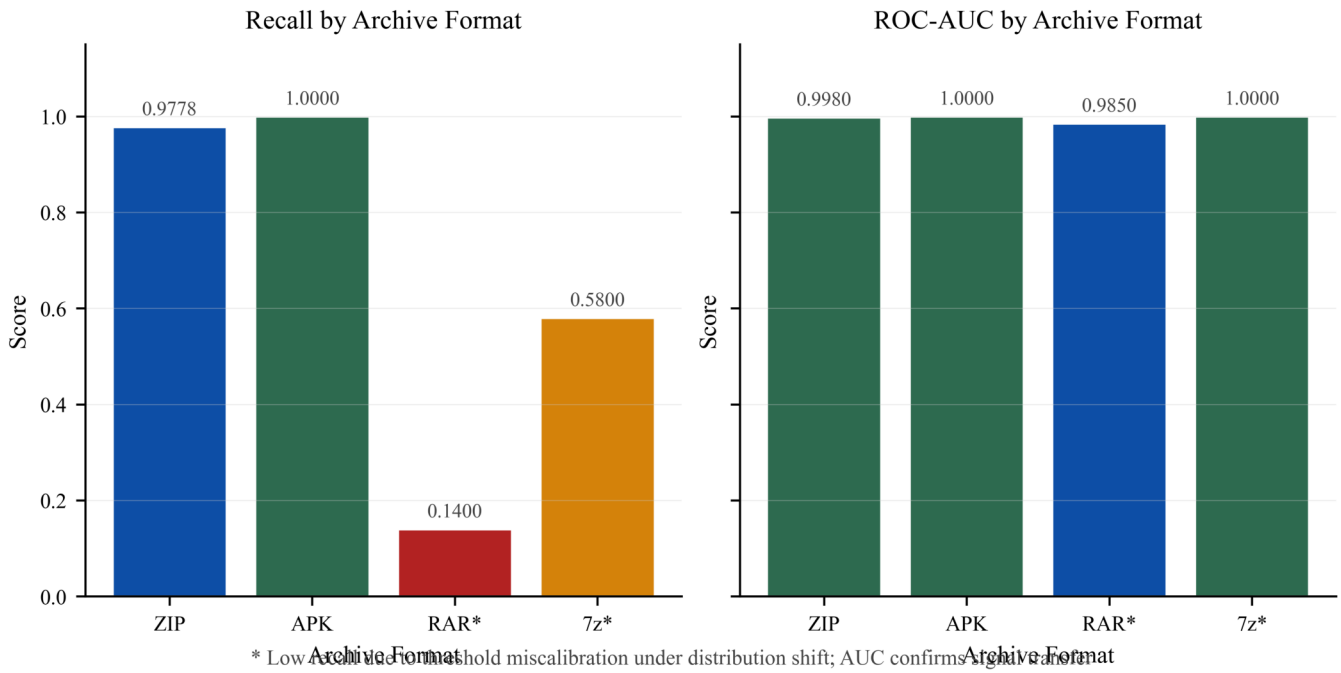

Table 1 - Baseline Comparison

table1_baseline_comparison.pdf

model	accuracy	precision	recall	f1	roc_auc	TP	FP	TN	FN
Rule-based baseli	0.8900	0.8792	0.8630	0.8710	-	233	32	325	37
mbieGuard XGB	0.9984	1.0000	0.9963	0.9981	1.0000	269	0	357	1

Table 2 - Dataset Statistics

table2_dataset_statistics.pdf

Component	Count	Derivation
Synthetic malicious files	1349	data/raw/malicious or label fallback
Synthetic benign files	1786	data/raw/benign or label fallback
Real-world validation files	1355	data/real_world_validation or label fallback
Processed malicious labels	1348	labels_df where label=1
Processed benign labels	1785	labels_df where label=0
Total processed samples	3133	len(labels_df)

Table 3 - Prevalence Breakdown

table3_prevalence_breakdown.pdf

Signal Type	Count	Share	Feature
Gootloader EOCD chaining (EOCD > 1)	68	5.0%	eocd_count
High entropy anomaly	24	1.8%	data_entropy_shannon
Undefined LFH method code	0	0.0%	lf_unknown_method
LFH/CDH mismatch	0	0.0%	method_mismatch
Total detected evasion	93	6.8%	label==1
Non-evasion / out-of-scope	1273	93.2%	label==0

Table 4 - Generalisation Results

table4_generalisation_results.pdf

Format	Model	Recall	Precision	F1	ROC-AUC	Notes
ZIP	XGBoost	0.9778	0.9962	0.9869	**0.9980**	
ZIP	Transformer	1.0000	1.0000	1.0000	**1.0000**	
APK	XGBoost	1.0000	0.9132	0.9547	**1.0000**	ZIP-based, full feature transfer
APK	Transformer	1.0000	1.0000	1.0000	**1.0000**	
RAR	XGBoost	0.1400	1.0000	0.2456	**0.9850**	Signal present, threshold miscalibration
RAR	Transformer	1.0000	0.5000	0.6667	**0.2031**	Flagged all samples as malicious
7z	XGBoost	0.5800	1.0000	0.7342	**1.0000**	
7z	Transformer	1.0000	0.5000	0.6667	**0.2083**	Flagged all samples as malicious
RAR (t=0.15)	XGBoost-calibrated	0.3600	1.0000	0.5294	**0.9850**	Signal present, threshold miscalibration
7z (t=0.25)	XGBoost-calibrated	0.7650	1.0000	0.8669	**1.0000**	